

Introduction to Enamel

- Enamel is a **unique calcified tissue**
- **Ectodermal origin** (from enamel epithelium)
- Covers the **crown of the tooth**
- Most **highly mineralized tissue in the body**
- Contains:
 - Very little organic material
 - No cells → **non-living tissue**

Healthy Enamel

Demineralized Enamel



Carbonated
hydroxyapatite
enamel crystal

Demineralization

Acid in plaque

Partly dissolved
crystal

Remineralization

Fluoride in plaque

Fluorapatite-like
coating on remineralized
crystal



Enamel Characteristics

- Enamel is the outermost protective layer of the tooth crown.
- It is derived from ectoderm via ameloblasts.
- It is acellular, avascular, and non-vital after formation.
- Exhibits highest degree of mineralization among body tissues.
- Formed via epithelial-mesenchymal interactions during tooth development.

Enamel Junctions

- Enamel meets dentine at:
 - **Enamel–Dentine Junction (EDJ)**
- Other important junctions:
 - Cementoenamel junction (CEJ)
 - Cementodentine juncti

Histological Appearance

- Depends on preparation method:
 - **Ground sections** → enamel visible
 - **Decalcified sections** → enamel lost
- Reason:
 - Enamel is almost entirely **mineral**

Histological Considerations

- Ground sections preserve mineralized enamel structure.
- Decalcified sections remove enamel due to high mineral content.
- Optical properties vary due to refractive index differences.
- Enamel appears birefringent under polarized light.
- Structural study mainly relies on ground sections.

Physical Properties of Enamel

- Hardest biological tissue
- High **abrasion resistance**
- Brittle but:
 - Supported by dentine → prevents fracture
- High **elastic modulus**
- Specific gravity ≈ 3

TABLE 12-2

Enamel Structure Loss

METHOD	FEATURES	CLINICAL APPEARANCE
Attrition	Loss through tooth-to-tooth contact from mastication with aging or more severely with parafunctional habits	• Matching wear on masticatory surfaces • Shiny facets on amalgam contacts • Enamel and dentin wear at the same rate • Possible fracture of cusps or restorations
Abrasion	Loss through friction from tooth-brushing and/or toothpaste	• Usually located at facial cervical regions • Lesions more wide than deep • Canines commonly affected because of tooth position
Erosion	Loss through chemical means (via acid) not involving bacteria	• Broad concavities within smooth surface enamel • Cupping of occlusal surface (or incisal grooving) with dentin exposure (with possible dentinal hypersensitivity) • Increased incisal translucency • Wear on nonmasticatory surfaces (exact location depends on acid intake-type) • Raised and shiny amalgam restorations • Preservation of enamel cuff in gingival crevice • Pulp exposure and loss of surface characteristics of enamel in primary teeth
Caries	Loss through chemical means (via acid) from cariogenic bacteria by way of dental biofilm	• All surfaces can be affected • Occlusal surfaces more commonly affected, especially in pits and grooves • Possibly rapid progression of interproximal smooth surface lesions if progress goes unchecked • Cervical lesions sometimes secondary to other forms of hard tissue loss or gingival recession
Abfraction	Possible loss through tensile and compressive forces during tooth flexure with parafunctional habits	• Can affect both facial and lingual cervical regions • Deep, narrow V-shaped notch • Commonly affects single teeth that have occlusal loads

Physical Properties

- Hardest biological tissue with high compressive strength.
- Brittle in nature but supported by resilient dentine.
- High modulus of elasticity reduces fracture risk.
- Surface enamel is more mineralized than subsurface.
- Shows gradual variation in hardness from surface to EDJ.

Thickness of Enamel

- Thickest at:
 - Cusps & incisal edges (~2.5 mm)
- Thinnest at:
 - Cervical margin
- Varies:
 - Between teeth
 - Between individuals

Composition of Enamel

Component	Percentage
Mineral (Hydroxyapatite)	95–96%
Water	~2%
Organic Matrix	1–2%

Amelogenesis (Flowchart)

Dental Lamina → Tooth Bud → Cap Stage → Bell Stage

→ Differentiation of Ameloblasts

→ Matrix Secretion (Amelogenins)

→ Mineralization Begins

→ Maturation Phase (Protein removal + crystal growth)

→ Fully formed Enamel

Enamel Prisms

- Basic structural unit of enamel.
- Extend from enamel-dentine junction to surface.
- Composed of tightly packed hydroxyapatite crystals.
- Show keyhole pattern (head and tail regions).
- Orientation changes provide mechanical strength.

Hunter-Schreger Bands

- Alternating light and dark bands seen in enamel.
- Caused by periodic changes in prism direction.
- Improve resistance to crack propagation.
- Composed of parazones and diazones.
- Prominent in inner enamel.

Clinical Significance

- Enamel cannot regenerate once damaged.
- Susceptible to caries and erosion.
- Fluoride incorporation enhances resistance.
- Demineralization-remineralization balance is critical.
- Basis for preventive dentistry strategies.